Automotive Engineering MSc: Group Design Project 2025

HydRover – Team 1
Powertrain Department

Design of a Retrofittable Hydrogen Hybrid Powertrain for an Off-Road Vehicle

Automotive Group Design Project Presentation Day, Wednesday 30 April 2025

Scope and Objectives

Design a cost-effective, retrofittable Hydrogen ICE powertrain for a Gasoline L322 Range Rover that maximises performance whilst eliminating CO_2 emissions

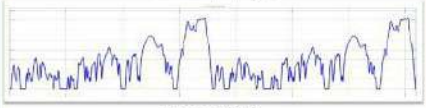
Use Case: Urban Driving

- Use case based on **survey data**
- Frequent stops
- Varying speeds
- Short-distance trips (<240 km per week)
- **Target Regions:** Germany, Switzerland
  → High hydrogen refuelling density

Range Estimation

WLTP Drive Cycle

Velocity (m/s)



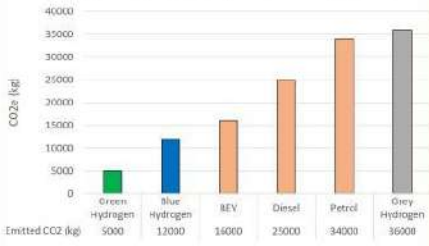
Power (W)

Simulates real-world **urban driving conditions**:

- Frequent starts and stops
- Varying acceleration and deceleration
- Mix of low to moderate speeds
- Occasional high-speed segments

Emissions

Life cycle CO_2 emissions by fuel after 100,000 km



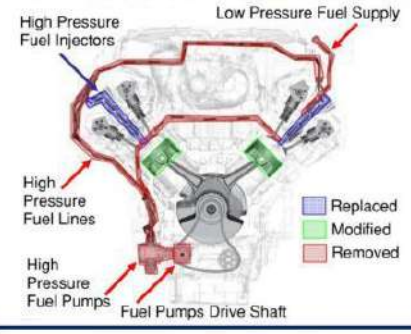
- **2022**: 92% of produced Hydrogen is Grey
- **2035**: 22% of will be Green or Blue
- **H_2 consumption:** 2.6 kg per 100 km

Costs

Base Powertrain ICE Hydrogen	Upgraded Powertrain ICE Hydrogen + MHEV
£16,000 - £19,000	£24,000 - £27,000

- Price above target market expectations
  → Customers expect to pay under £9000

5.0L V8 Supercharged Engine From Gasoline to Hydrogen ICE



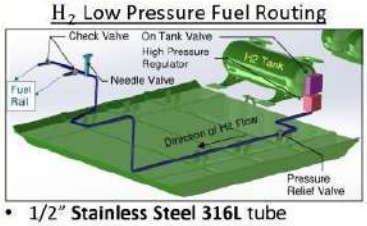
Engine Performance

- **Fuel injection strategy** optimised for high efficiency and low NOx emissions using **late injection** for stratified mixture
- Achieves **Euro 7** NOx compliance and zero CO_2 emissions
- Increase in peak efficiency compared to the original engine due to hydrogen's wide flammability range and high flame speed

Hydrogen Engine Peak Efficiency at 85 bars Injection Pressure	Hydrogen Engine Peak Efficiency at 40 bars Injection Pressure
44%	40%

System Modifications

H_2 Low Pressure Fuel Routing



- 1/2" **Stainless Steel 316L** tube
- Max rated pressure: 310 bars
- Operating pressure: 45 bars
- **FOS = 6.8**; SAE J2579 minimum = 1.8

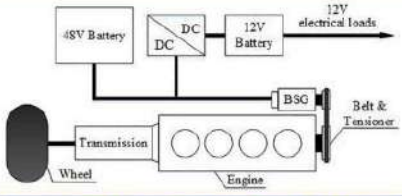
Nickel Alloy Spark plugs

- Cold-rated non-platinum plugs
- Prevents unintended ignition
- Designed for **lean-burn** applications

Phinia H_2 injectors & Bosch Fuel Rail

- Designed for direct injection
- Accurate, repeatable delivery
- Nominal pressure range **up to 40 bars**

Hybridisation – 48V Mild Hybrid



Range without Battery (km)	Range with Battery (km)
370	411

Belt Starter Generator

- Valeo 48V iBSG
- Peak Power: 10kW
- Peak Regen: 12kW

Battery

- BOSCH Standard
- Voltage: 48V
- Capacity: 360Wh

- **Realistic model** using WLTP cycle with dynamic fuel use, power split, regenerative braking and SOC feedback
- **Optional hybridisation upgrade** to keep the base model affordable
- **Start and stop** functionality
- **20% torque assist**
- **Range extension**
- Immediate energy use for best battery sizing strategy
- 12V Generator replaced by the 48V iBSG

Conclusion

- **TRL 2**: Technology concept formulated
- Retrofitting is possible but **highly constrained**
- **Compromises:** Limited tank space, vehicle mass
- **Opportunities:** Emissions cut, fleet reuse
- **Challenges**: Cost and H_2 infrastructures

Team Members

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